WORKSHEET

Chapter 7: Using spreadsheets to calculate interest

In this worksheet, you are going to create a spreadsheet to make interest calculations for investments using both simple and compound interest. The spreadsheet can be used to make comparisons between different investments to determine which yields the most interest.

1 Copy the data and formulas below into a spreadsheet. It is very important to copy them exactly, especially the formula cells (the ones that start with an '=' sign).

	A	B	C	D	E	F
1						
2		Principle				
3		Interest rate per period, %				
4		Number of periods				
5						
6						
7						
8		**Simple Interest Calculation**			**Compound Interest Calculation**	
9		*I = Prn /100*			*A=P(1+r/100)^n*	
10		Interest	=C2*C3*C4/100		Interest	=F11-C2
11		Total Value of Investment	=C2+C10		Total Value of Investment	=C2*(1+C3/100)^C4

Tip: See if you can format cells C2, C10, C11, F10 and F11 as Accounting, with a $ sign and two decimal places. This won't change your answers, but will make the spreadsheet easier to read.

2 Cells C2, C3 and C4 must be filled in with the data for each particular question. For example, if we were calculating the value of an investment of $100 000 at 7.5% p.a. interest for 1 year and comparing the results for both simple and compound interest, we would fill them in as shown below.

	A	B	C	D	E	F
1						
2		Principle	$100,000.00			
3		Interest rate per period, %	7.5			
4		Number of periods	1			

a Add the above data to your spreadsheet.

b What do you notice about the total value of the investment with simple interest when compared to compound interest?

c How much interest is earned with each form of investment?

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3 a Use the spreadsheet to help you fill in the table below.

Principle ($)	Interest rate per period (%)	No. of periods	Simple interest ($)		Compound interest ($)	
			Interest earned	Value of investment	Interest earned	Value of investment
100 000	7.5	1				
100 000	7.5	2				
100 000	7.5	4				
100 000	7.5	7				
100 000	7.5	10				
100 000	7.5	50				

b What do you notice about the value of the two different investments as the number of periods increases?

c How much more interest is earned over 50 years when interest is compounded annually than when it is calculated as simple interest?

4 Use the spreadsheet to fill in the following table.

Principle ($)	Interest rate per period (%)	No. of periods	Simple interest ($)		Compound interest ($)	
			Interest earned	Value of investment	Interest earned	Value of investment
100	12.00	5				
5000	8.75	3				
24 000	3.21	10				
30 000	10.00	7				
75 000	4.25	6				

5 When interest is compounded more than once each year, we need to adjust the values for Interest rate per period and No. of periods. Complete the following table, giving values to three decimal places.

	Interest rate for different time periods (%)			
Interest rate per annum (%)	Interest rate per 6 months	Quarterly interest rate	Monthly interest rate	Daily interest rate
8.00				
5.00				
12.00				
3.25				
7.80				

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6 Use the spreadsheet to calculate the total value of the following investments. Don't forget to adjust the interest rate per period and the number of periods for each question.

a $5000 invested for 8 years at 12% p.a., compounded quarterly.

b $2300 invested for 5 years at 3.5% p.a., compounded every 6 months.

c $28 000 invested for 10 years at 14.6% p.a., compounded monthly.

d $1000 invested for 6.3% p.a., compounded daily for 20 years.

7 Ruby has $40 000 to invest. She is choosing between the following term-deposit accounts.

A Investment Maker: 6.7% p.a. simple interest for 10 years.

B Incentive Saver: 5.4% p.a. compounded half-yearly for 10 years.

C Money Spinner: 5.2% p.a. compounded quarterly for 10 years.

D Super Saver: 4.8% p.a. compounded monthly for 10 years.

a Use the spreadsheet to determine how much interest Ruby would earn in each account.

b Which account would be the best choice for Ruby?

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Answers

2 **b** They are the same: $107 500.

c $7500

3 **a**

Principle ($)	Interest rate per period (%)	No. of periods	Simple interest ($)		Compound interest ($)	
			Interest earned	Value of investment	Interest earned	Value of investment
100 000	7.5	1	7500	107 500	7500.00	107 500.00
100 000	7.5	2	15 000	115 000	15 562.50	115 562.50
100 000	7.5	4	30 000	130 000	33 546.91	113 546.91
100 000	7.5	7	52 500	152 500	65 904.91	165 904.91
100 000	7.5	10	75 000	175 000	106 103.16	206 103.16
100 000	7.5	50	375 000	475 000	3 618 974.60	3 718 974.60

b They both increase, but the investment with compound interest increases at a much greater rate.

c $3 243 974.60

4

Principle ($)	Interest rate per period (%)	No. of periods	Simple interest ($)		Compound interest ($)	
			Interest earned	Value of investment	Interest earned	Value of investment
100	12.00	5	60.00	160.00	76.23	176.23
5000	8.75	3	1312.50	6312.50	1430.69	6430.69
24 000	3.21	10	7704.00	31 704.00	8917.67	32 917.67
30 000	10.00	7	21 000.00	51 000.00	28 461.51	58 461.51
75 000	4.25	6	19 125.00	94 125.00	21 275.91	96 275.91

5

	Interest rate for different time periods (%)			
Interest rate per annum (%)	Interest rate per 6 months	Interest rate per annum	Interest rate per 6 months	Interest rate per annum
8.00	4.000	2.000	0.667	0.022
5.00	2.500	1.250	0.417	0.014
12.00	6.000	3.000	1.000	0.033
3.25	1.625	0.813	0.271	0.009
7.80	3.900	1.950	0.650	0.021

6 **a** $12 875.41

b $2735.72

c $119 509.30

d $3525.04

7 **a** A $26 800; B $28 150.47; C $27 056.03; D $24 581.11

b B Incentive Saver

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